https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Chapter 14

Introduction to Wireless Networking

A huge part of networking today is working with wireless networks. When preparing for your ICND1 exam, it is important to have the basics of wireless networking down, including wireless standards and security best practices. This chapter helps prepare you for any wireless networking questions you may get by testing you on wireless basics, standards, and protocols.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding wireless networking basics, such as the types of wireless networks
- Understanding wireless standards and their characteristics
- Getting familiar with wireless networking security best practices
- Handling wireless connectivity issues

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- A number of different agencies deal with wireless as a standard. The Federal Communications Commission (FCC) is responsible for regulating the use of wireless devices and frequencies, whereas the Institute of Electrical and Electronics Engineers (IEEE) is responsible for defining the wireless standards. The Wi-Fi Alliance is responsible for ensuring compatibility and interoperability of wireless networking components.
- Some types of wireless networks, such as ad-hoc wireless networks, involve wireless clients connecting to one another without the use of an access point. This is also known as an *independent basic service set (IBSS)*. Using a wireless access point to connect all your wireless devices means you are running wireless in infrastructure mode, also known as a *basic service set (BSS)*. Using multiple access points to allow users to roam throughout the building is known as an *extended service set (ESS)*.

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

Wireless Basics

504–524 Choose the best answer(s).

504. What type of wireless network involves two wireless clients connecting to one another?

(A) FCC

(B) Infrastructure

(C) Domain-based

(D) Ad-hoc

505. Which of the following is considered the name of the wireless network?

(A) ESS

(B) SSID

(C) TKIP

(D) FCC

506. Which agency is responsible for defining the wireless standards?

(A) Wi-Fi Alliance

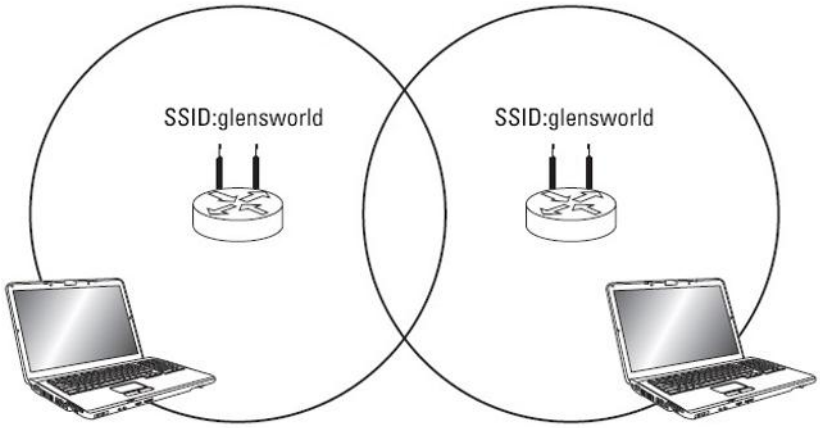
(B) FCC

(C) IETF

(D) IEEE

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

507. In the figure shown below, what type of wireless network exists?



(A) ESS

(B) BSS

(C) TKIP

(D) FCC

508. Which of the following devices could cause interference with your wireless network? (Select two.)

(A) TV remote

(B) Cordless phone

(C) Microwave

(D) Toaster

509. You are installing a wireless network for a small office environment. The manager is worried that someone in the parking lot can connect to the wireless network. What could you do to help minimize the chances of that happening?

(A) Change the SSID.

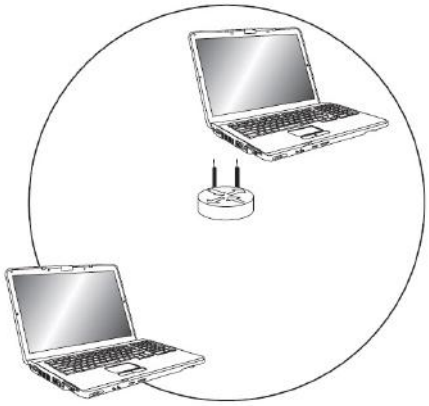
(B) Change the ESSID.

(C) Enable SSID broadcasting.

(D) Change the power levels on the access point.

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

510. In the figure shown below what type of wireless network exists?



(A) ESS

(B) TKIP

(C) BSS

(D) FCC

511. You are experiencing intermittent problems with your wireless network and you suspect it is interference from other wireless devices. What can you do to solve the problem?

(A) Change the SSID.

(B) Enable MAC filtering.

(C) Disable SSID broadcasting.

(D) Change the channel on the wireless access point.

512. What type of wireless network involves the wireless clients connecting to a wireless access point to access the network?

(A) FCC

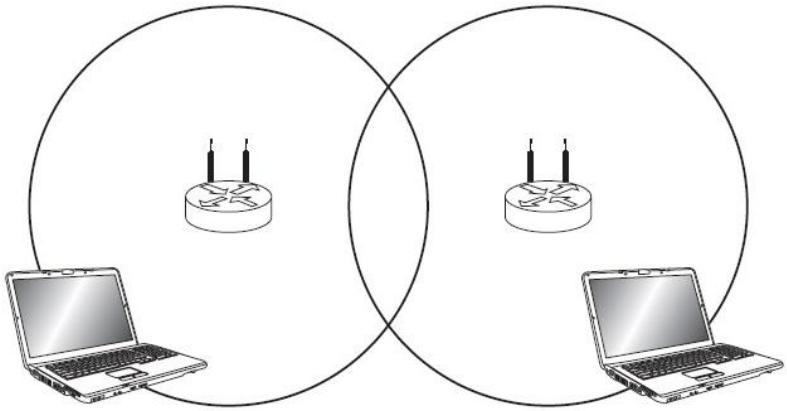
(B) Infrastructure

(C) Domain-based

(D) Ad-hoc

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

513. Which of the following facts are true of the WLAN diagram displayed below? (Select two.)



(A) The diagram represents an ESS network with each access point using a different SSID.
(B) The diagram represents an ESS network with both access points using the same SSID.
(C) The diagram represents a BSS network with both access points using the same SSID.
(D) Each access point is using a different channel.
(E) Each access point is using the same channel.

514. Which of the following are requirements for an ESS network? (Select three.)
(A) Use multiple access points with different SSIDs.
(B) Use multiple access points with the same SSIDs.
(C) Each access point uses a different frequency range.
(D) Each access point uses the same frequency range.
(E) Access points must overlap by 10 percent or greater.
(F) Access points must overlap by less than 10 percent.

515. Which agency is responsible for regulating the use of wireless and wireless frequencies?
(A) Wi-Fi Alliance
(B) IEEE
(C) IETF
(D) FCC

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

516. Which of the following is another term for an ad-hoc wireless network?

(A) BSS

(B) IBSS

(C) ESS

(D) FCC

517. You are the administrator for a small office and have configured four access points to create the WLAN. What type of WLAN is created?

(A) BSS

(B) ESS

(C) IBSS

(D) FCC

518. Which type of wireless network uses a single wireless access point with an SSID assigned?

(A) BSS

(B) ESS

(C) TKIP

(D) FCC

519. You have manufactured your own wireless network card and would like it tested and certified for Wi-Fi compliance. What organization would do this?

(A) Wi-Fi Alliance

(B) IEEE

(C) IETF

(D) FCC

520. You have installed a single access point to create a WLAN for a small office. What type of WLAN did you create?

(A) BSS

(B) ESS

(C) TKIP

(D) FCC

521. Which wireless LAN design ensures that a roaming user does not loose connectivity when moving from one wireless access point to another?

(A) Ensure that all access points are using the same channel.

(B) Ensure that each access point has a different SSID.

(C) Ensure that the manufacturer of the access point and wireless network cards are the same.

(D) Ensure that the area of coverage by both access points overlaps by at least 10%.

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

522. Which agency is responsible for ensuring compatibility with wireless networking components?

(A) Wi-Fi Alliance

(B) IEEE

(C) IETF

(D) FCC

523. Which three wireless channels do not overlap with one another?

(A) 2, 4, 8

(B) 1, 6, 11

(C) 4, 8, 9

(D) 3, 6, 9

524. Which type of wireless network supports roaming users by having multiple access points share the same SSID?

(A) BSS

(B) ESS

(C) TKIP

(D) FCC

Wireless Security

525–535 Choose the best answer(s).

525. Which of the following should be done with the SSID on a wireless network in order to follow good security practices? (Select two.)

(A) Change the SSID.

(B) Keep the default SSID.

(C) Enable SSID broadcasting.

(D) Disable SSID broadcasting.

526. Which of the following represent good security practices for wireless networks? (Select three.)

(A) Disable WPA2.

(B) Enable WPA2.

(C) Disable MAC filtering.

(D) Enable MAC filtering.

(E) Enable SSID broadcasting.

(F) Disable SSID broadcasting.

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

527. Which of the following wireless encryption protocols is an older protocol using RC4?

(A) WEP

(B) WAP

(C) WPA

(D) AES

528. Which of the following can be used to restrict which devices can connect to your wireless network?

(A) Change the SSID

(B) MAC Filtering

(C) Enable SSID broadcasting

(D) WPA2 encryption

529. You are installing a wireless access point for a small office. At a minimum, which wireless parameter must be configured in order for clients to connect to the wireless network?

(A) MAC filtering

(B) Port forwarding

(C) SSID

(D) WPA2 key

530. You are configuring a wireless network for a customer. What reason would you give the customer as to why WPA should be used instead of WEP for the encryption?

(A) WPA rotates the key with each packet.

(B) WPA forces the use of complex passwords.

(C) WPA uses a 24-bit IV.

(D) WPA keys stay the same the entire time the client is connected.

531. Which WPA2 mode involves using a pre-shared key as the method of security?

(A) Personal

(B) Enterprise

(C) ESK

(D) BSS

532. Which wireless encryption protocol uses the TKIP protocol?

(A) WEP

(B) WAP

(C) WPA

(D) AES

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

533. Which WPA mode uses a RADIUS server for authentication?

(A) Personal

(B) Enterprise

(C) WPA-PSK

(D) BSS

534. Which wireless encryption protocol uses AES for encryption?

(A) WEP

(B) WPA2

(C) WPA

(D) TKIP

535. What method of encryption does WPA2 use?

(A) RC4

(B) RC2

(C) AES-CCMP

(D) MD5

Wireless Standards

536–541 Choose the best answer(s).

536. Which wireless standard delivers data at 11 Mbps within the 2.4 GHz frequency range?

(A) 802.11a

(B) 802.11b

(C) 802.11g

(D) 802.11n

537. Which wireless standard is capable of transferring data over 150 Mbps and runs at both the 2.4 GHz and 5 GHz frequencies?

(A) 802.11a

(B) 802.11b

(C) 802.11g

(D) 802.11n

https://gicovietnam.blogspot.com - Kho tổng hợp tài liệu và chia sẻ

538. Which wireless standard runs at 54 Mbps and is compatible with 802.11b?

(A) 802.11a

(B) 802.11x

(C) 802.11g

(D) 802.11n

539. Which wireless network standard runs at 54 Mbps and uses the 5 GHz frequency?

(A) 802.11a

(B) 802.11b

(C) 802.11g

(D) 802.11n

540. Which IEEE standard defines wireless networking?

(A) 802.3

(B) 802.4

(C) 802.5

(D) 802.11

541. Which wireless standard is compatible with 802.11a?

(A) 802.13

(B) 802.11b

(C) 802.11g

(D) 802.11n